



Model Interpretability

Ricky Tharrington

Agenda

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- Background
- Individual Conditional Expectations (ICE)
- Partial Dependence
- Locally-Interpretable Model-Agnostic Explanations (LIME)
- Shapley Values
- Resources



Background

Model Interpretability

Background

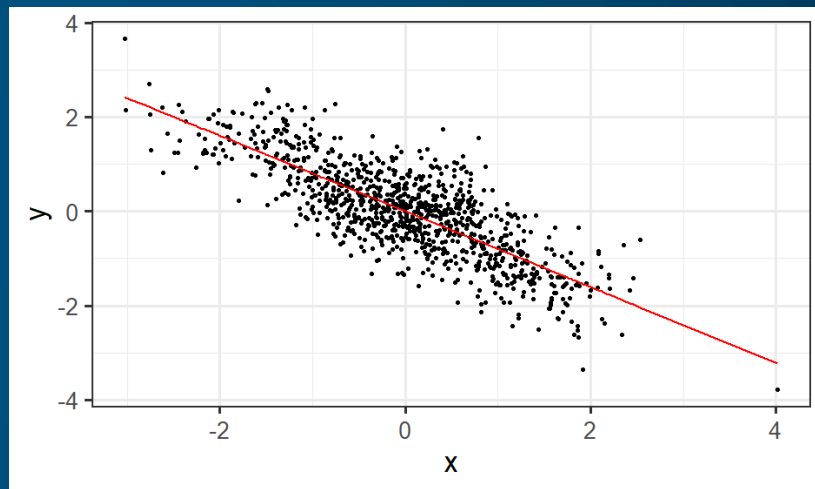
What is Model Interpretability?

- The ability for a human to understand a model's behavior
- Interpretations can be model and context dependent
- Answers a question
 - Why was this individual's loan application rejected?
 - Why is the symptom occurring in this patient?
 - Why is the stock price expected to go down?

Background

Linear Regression

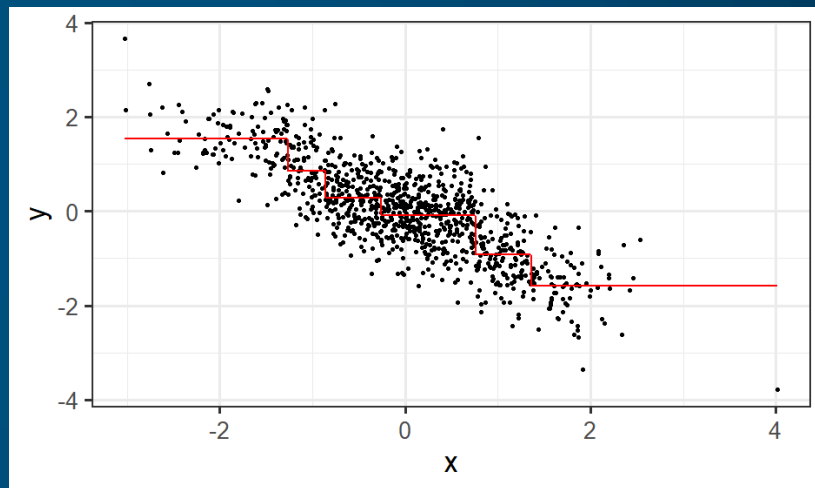
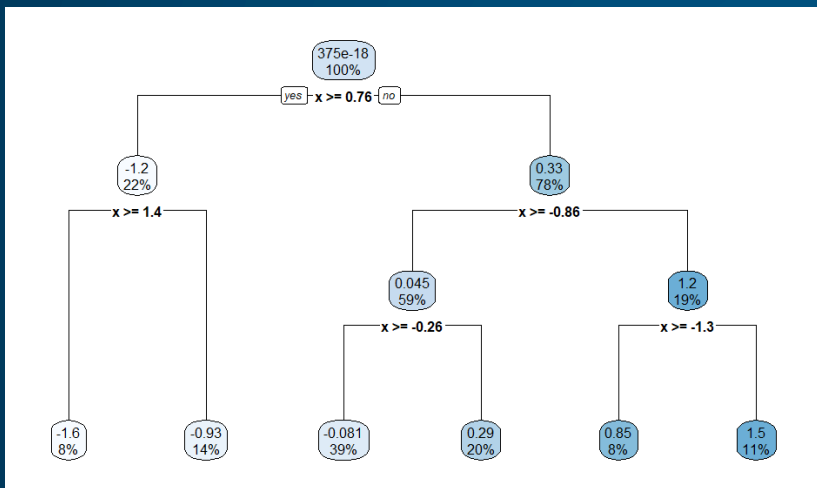
- $\hat{y} = \beta_0 + \beta_1 x$
- “As x increases by one, the expected value of y increases by β_1 .”



Background

Decision Trees

- “If X is greater than __ and less than __ then the expected value of y is...”

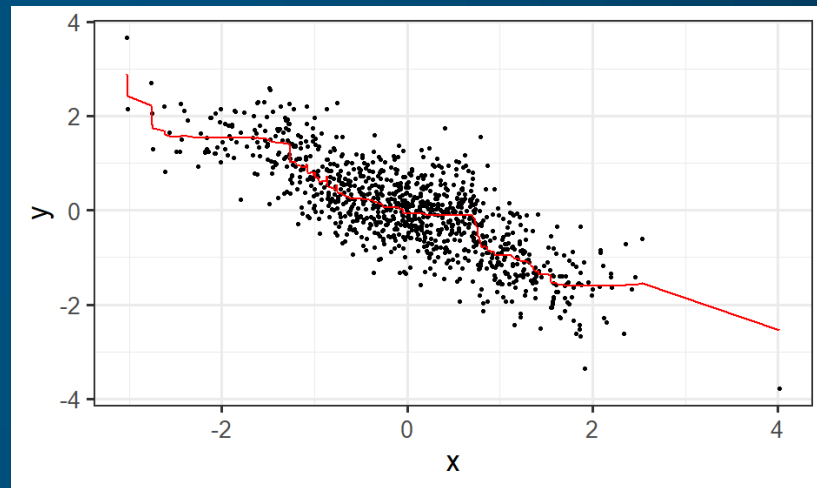


Background

Random Forests

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- “???”
- Interpretability is hopeless



Background

Importance of Interpretability

- Fairness / Transparency
 - Understanding a model improves consumer trust
 - Interpretations reveal model behavior on different classes
- Robustness
 - Interpretability methods can reveal overfitting issues
- Adverse Action notice requirements
 - Equal Credit Opportunity Act (ECOA)
 - Fair Credit Reporting Act (FCRA)

Background

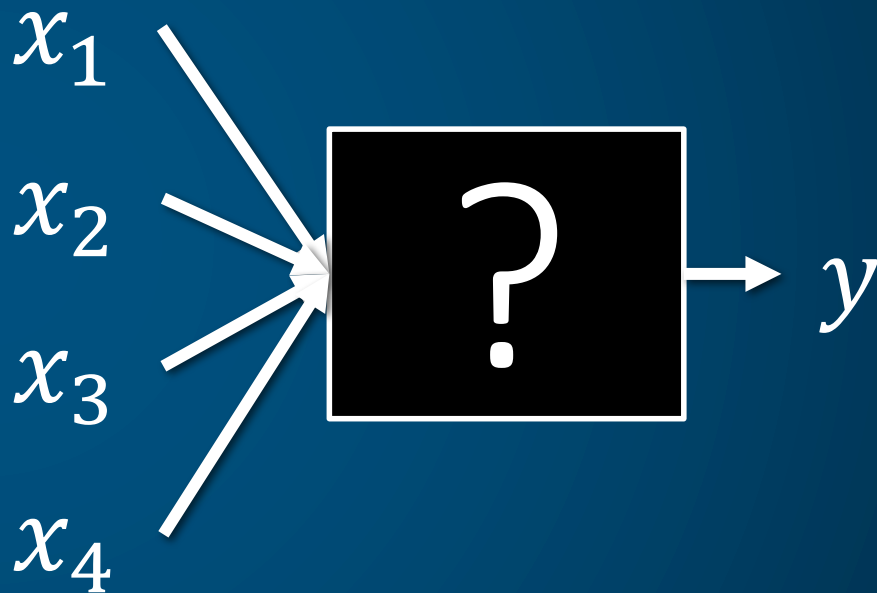
Flavors of Interpretability – Post Hoc

	Local	Global
Model Specific	Saliency Maps TreeSHAP	Tree Variable Importance
Model Agnostic	ICE LIME Shapley Values	Partial Dependence ALE

Background

Model-Agnostic Interpretability

- “Black Box” Interpretability
- Does not care about form of the model



Background

Methods in this Lecture

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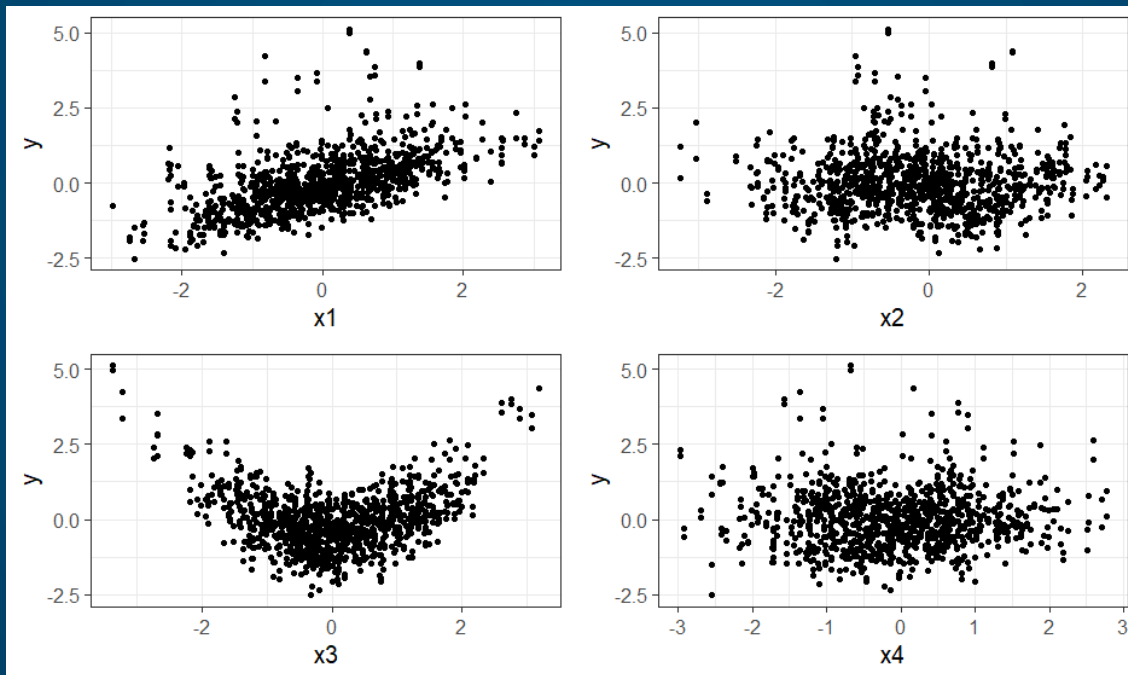
Method	Locality	Details
ICE	Local	Focuses on a single observation and a few variables
PD	Global	Uses all observations, a few variables
LIME	Local	Region around a single observation, linear model
Shapley	Local	Compares one observation to many

Background

Synthetic Data

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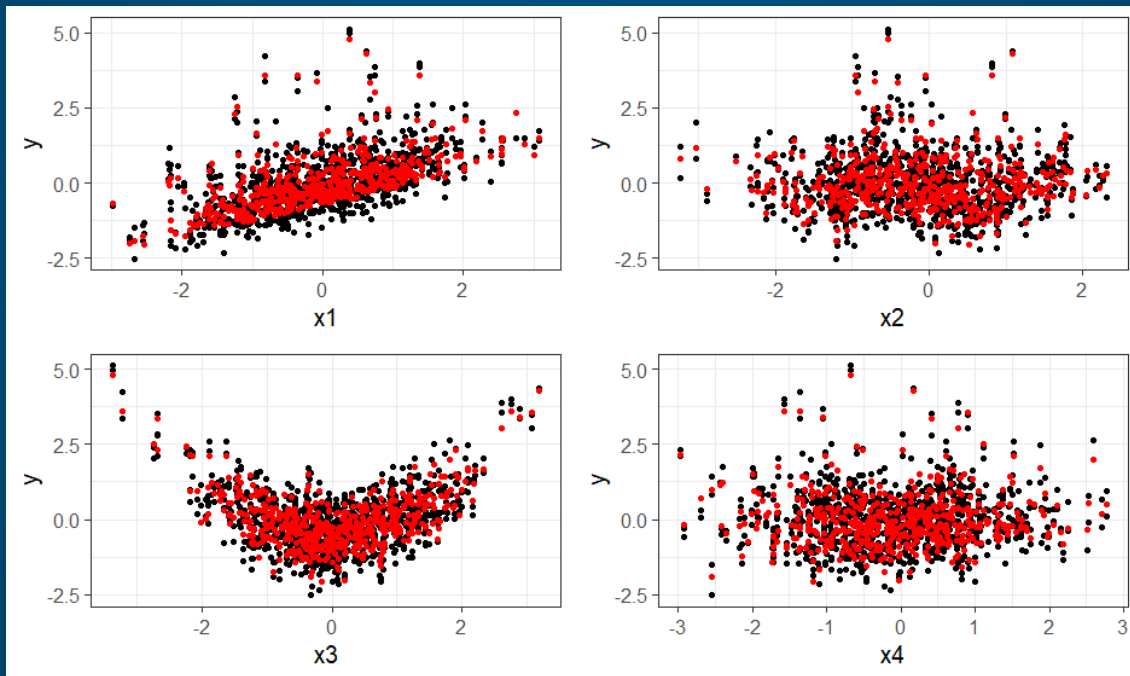
$$\hat{y} = 4x_1 + 2\sin \pi x_2 + 3x_3^2 + |x_4| + \varepsilon$$



Background

GAM Model

```
model = gam(y~x1+s(x2)+s(x3)+s(x4), data=sim_data)
```



Individual Conditional Expectations (ICE)

Model Interpretability

ICE

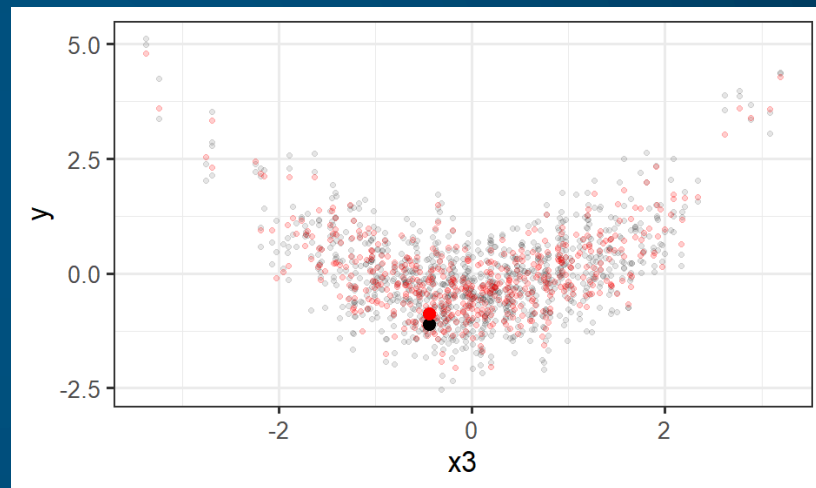
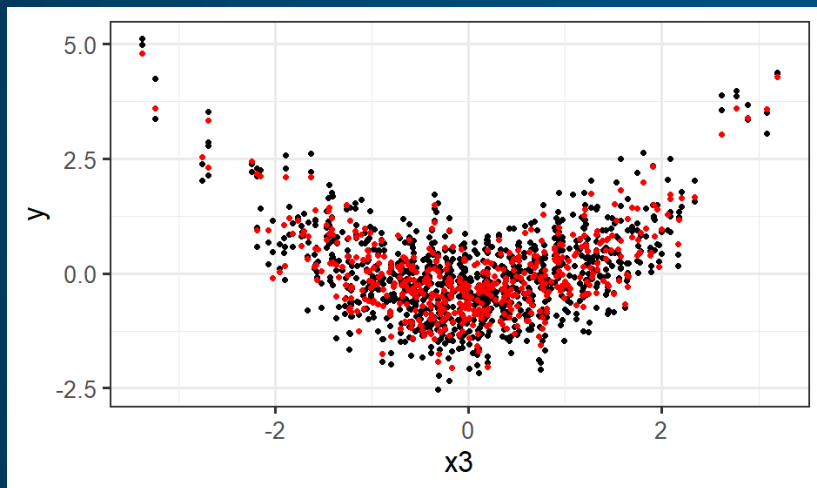
Details

- “Individual Conditional Expectations”
- Evaluates the effect on the model’s prediction of varying a variable’s value in a single observation
- Steps:
 1. Pick a single variable
 2. Pick a single observation
 3. Replicate observation, substitute range of values for variable
 4. Score new observations with model
 5. Plot
 6. Repeat for other observations

ICE

Single Variable, Single Observation

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ICE

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Pick an Observation

x1	x2	x3	x4
-0.44363	0.637281	-0.44017	-0.74696

ICE

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Vary the Variable of Interest (x3)

x1	x2	x3	x4
-0.44363	0.637281	-3	-0.74696
-0.44363	0.637281	-2	-0.74696
-0.44363	0.637281	-1	-0.74696
-0.44363	0.637281	0	-0.74696
-0.44363	0.637281	1	-0.74696
-0.44363	0.637281	2	-0.74696
-0.44363	0.637281	3	-0.74696

ICE

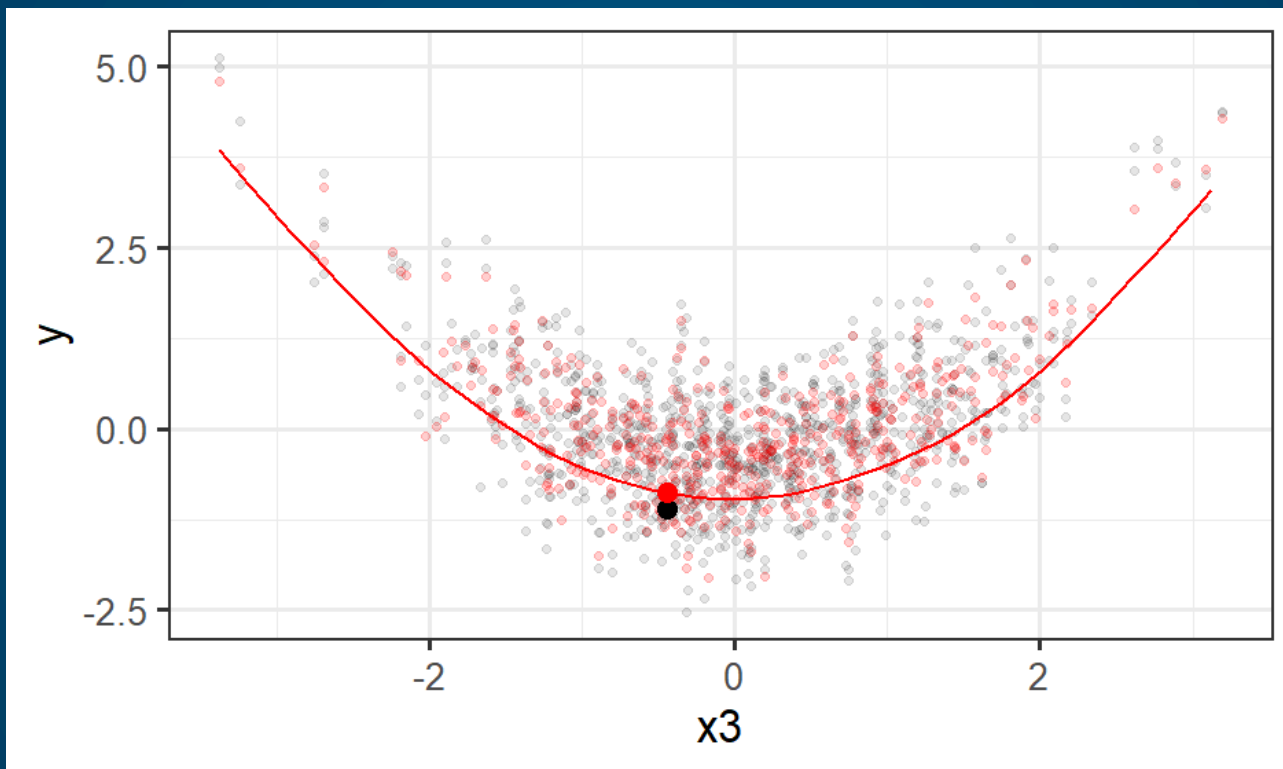
Score with Model

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x1	x2	x3	x4	yhat
-0.44363	0.637281	-3	-0.74696	2.937509
-0.44363	0.637281	-2	-0.74696	0.816968
-0.44363	0.637281	-1	-0.74696	-0.53431
-0.44363	0.637281	0	-0.74696	-0.9659
-0.44363	0.637281	1	-0.74696	-0.49077
-0.44363	0.637281	2	-0.74696	0.801018
-0.44363	0.637281	3	-0.74696	3.0092

ICE Plot

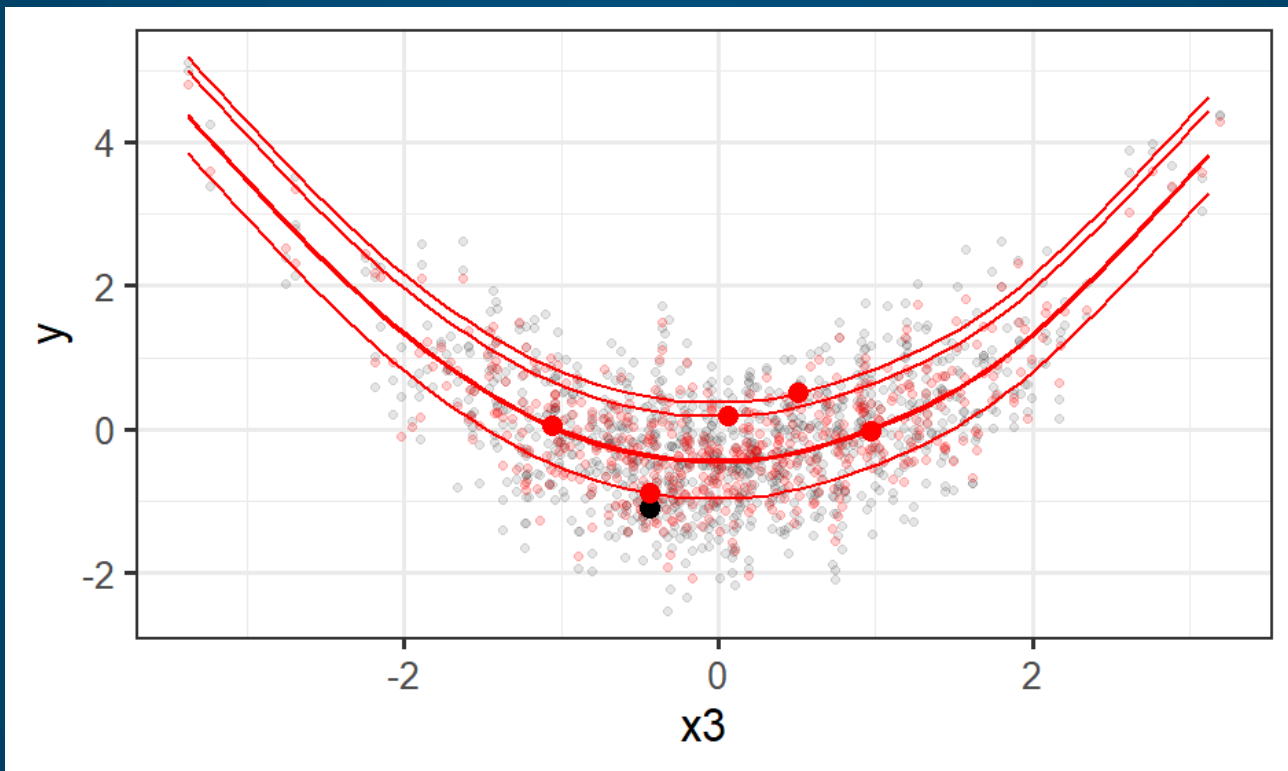
20



ICE

Repeat

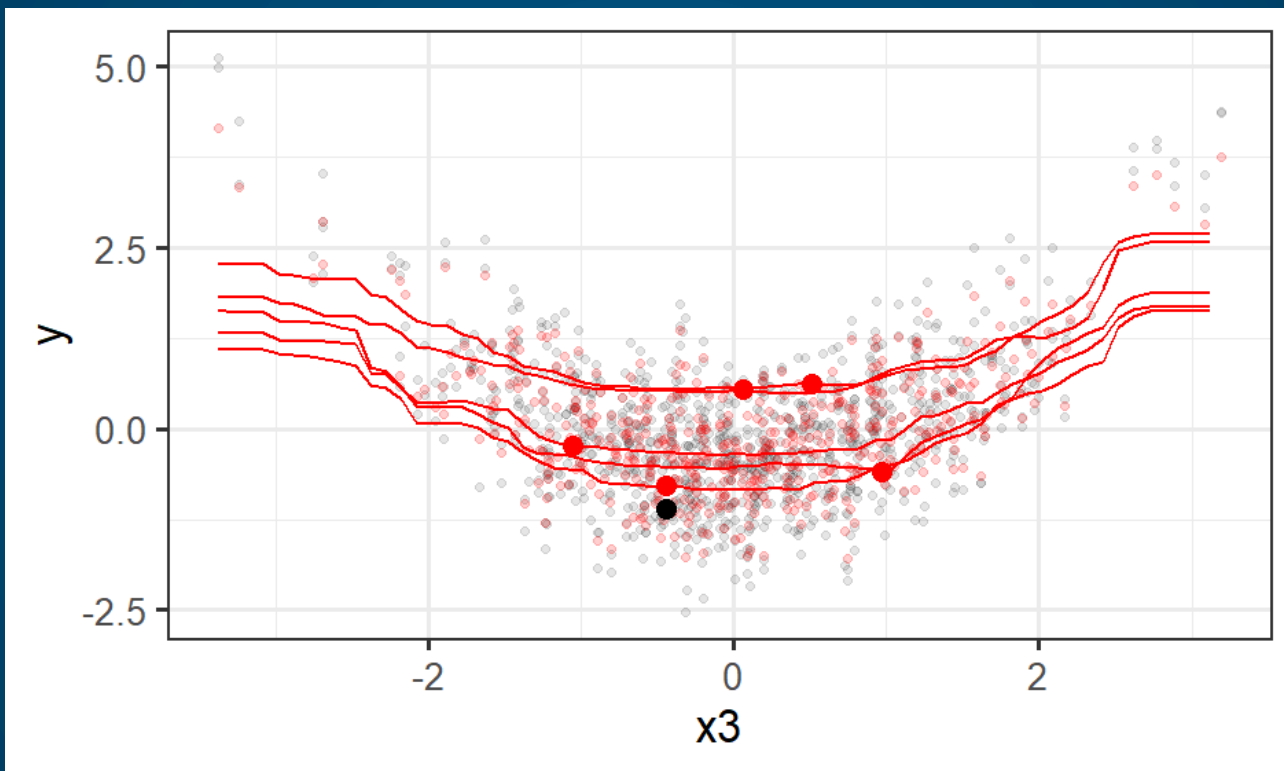
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ICE

Random Forest

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Partial Dependence

Model Interpretability

Partial Dependence

Details

- Evaluates the effect on the model's prediction of varying a variable's value in an entire dataset
- Steps:
 1. Pick a single variable
 2. Replicate dataset, substitute range of values for variable
 3. Score new observations with model
 4. Average prediction for each dataset replicate
 5. Plot
 6. Repeat for other variables

Partial Dependence

Dataset Replication

x1	x2	x3	x4
-0.44363	0.637281	-0.44017	-0.74696
-0.5148	-0.73199	0.969385	1.631754
1.180671	-0.74499	0.509752	1.047877
1.573187	0.264565	0.060567	-1.03539
0.09336	-2.04954	-1.05795	0.634159
0.165143	-0.33239	-0.69795	-2.01421
-0.33853	0.522166	0.364564	-0.63017

Partial Dependence

Dataset Replication – Replace Column

x1	x2	x3	x4
-0.44363	0.637281	0	-0.74696
-0.5148	-0.73199	0	1.631754
1.180671	-0.74499	0	1.047877
1.573187	0.264565	0	-1.03539
0.09336	-2.04954	0	0.634159
0.165143	-0.33239	0	-2.01421
-0.33853	0.522166	0	-0.63017

Partial Dependence

Dataset Replication – Score with Model

x1	x2	x3	x4	yhat
-0.44363	0.637281	0	-0.74696	-0.9659
-0.5148	-0.73199	0	1.631754	-0.46384
1.180671	-0.74499	0	1.047877	0.373391
1.573187	0.264565	0	-1.03539	0.182268
0.09336	-2.04954	0	0.634159	-0.43033
0.165143	-0.33239	0	-2.01421	-0.0278
-0.33853	0.522166	0	-0.63017	-0.94303

Partial Dependence

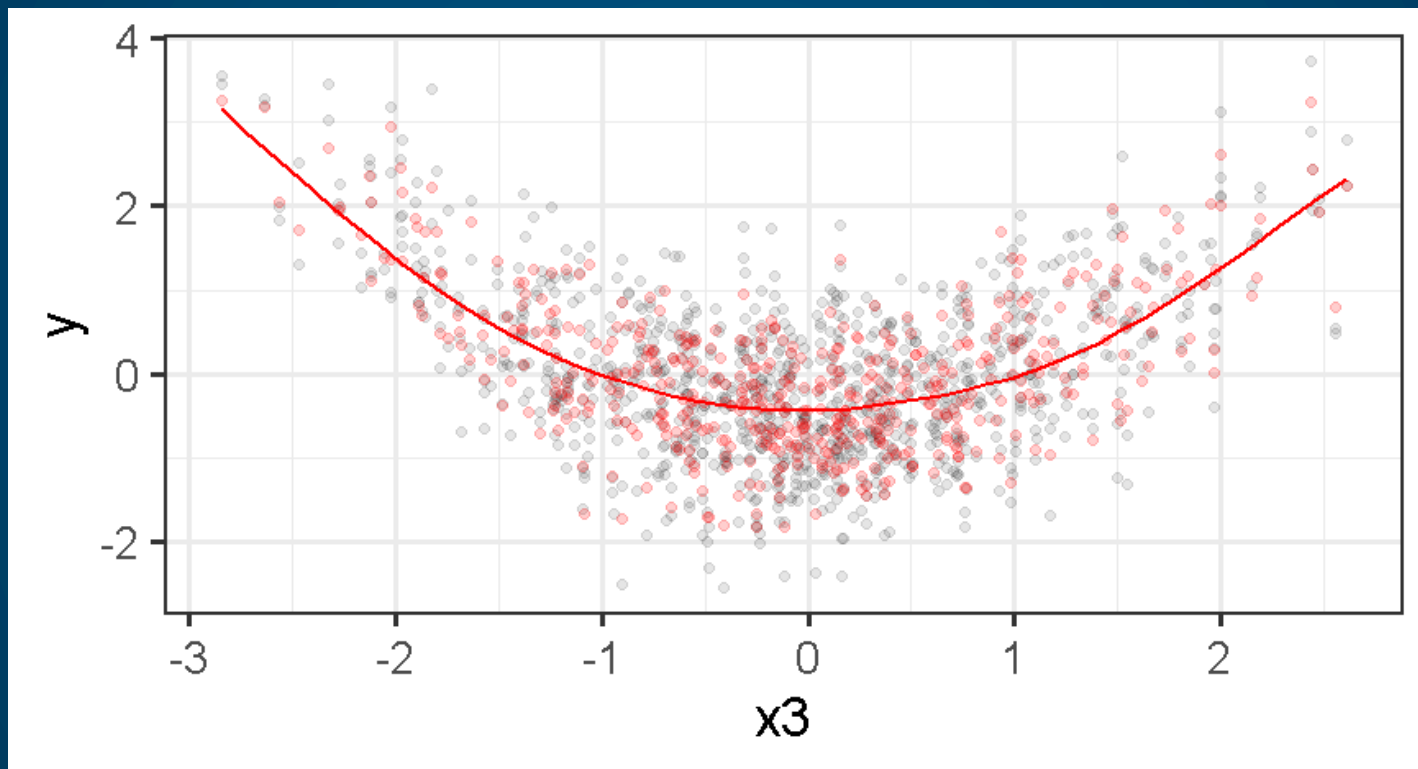
Aggregate for Every Copy

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x3		Mean Prediction
	-3	3.416149
	-2	1.295609
	-1	-0.05567
	0	-0.48726
	1	-0.01213
	2	1.279659
	3	3.487841

Partial Dependence Plot

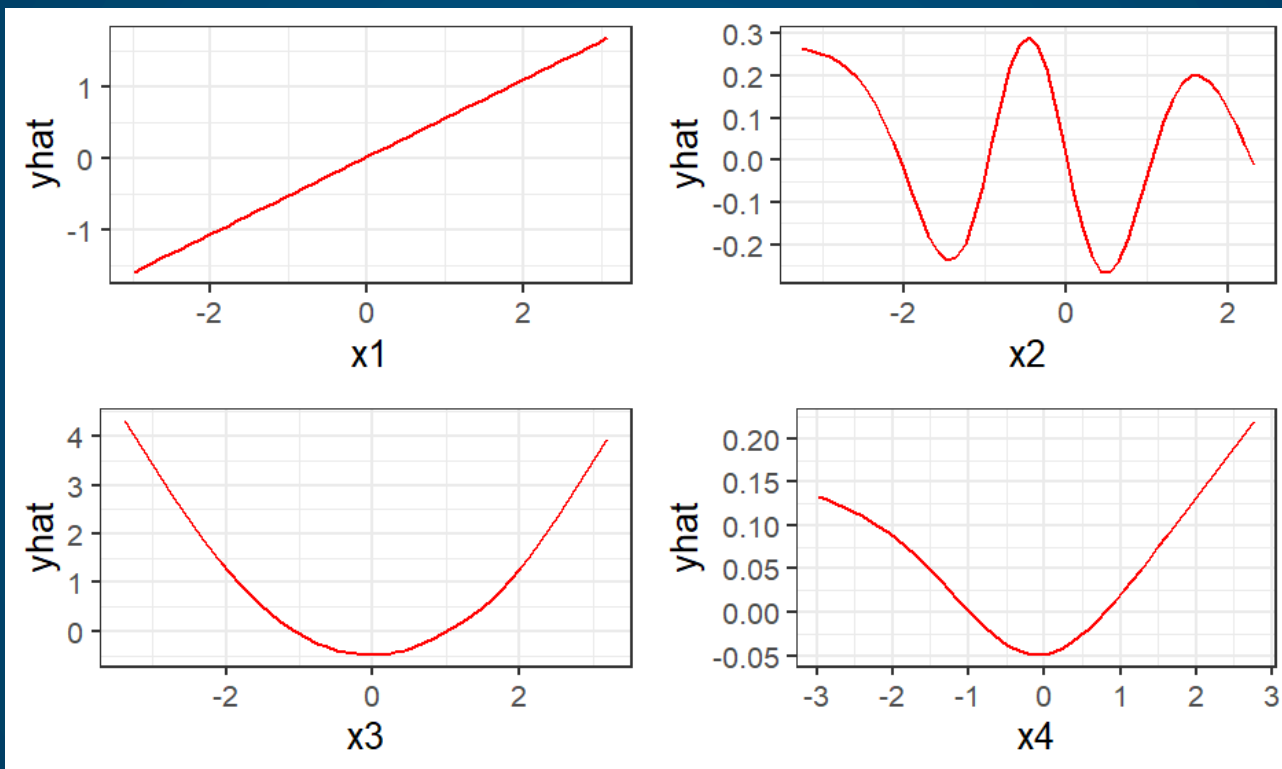
29



Partial Dependence

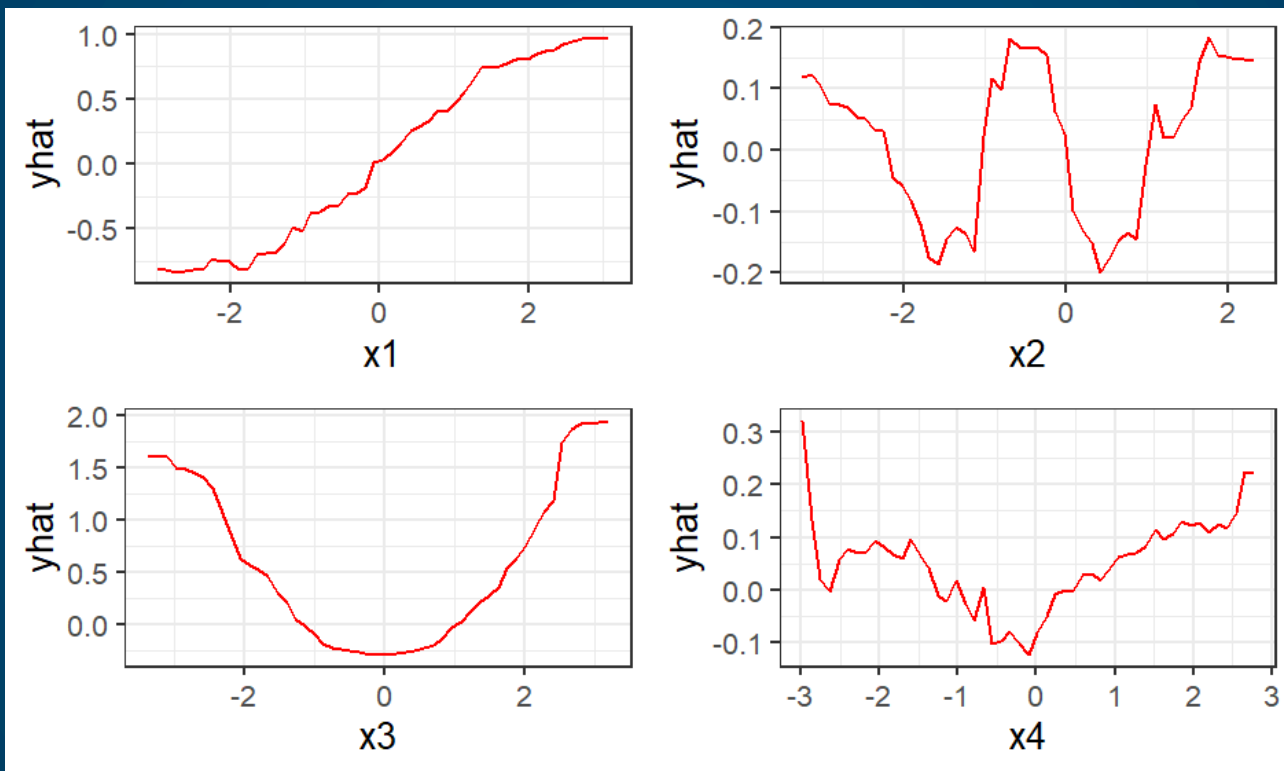
All Variables - GAM

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Partial Dependence

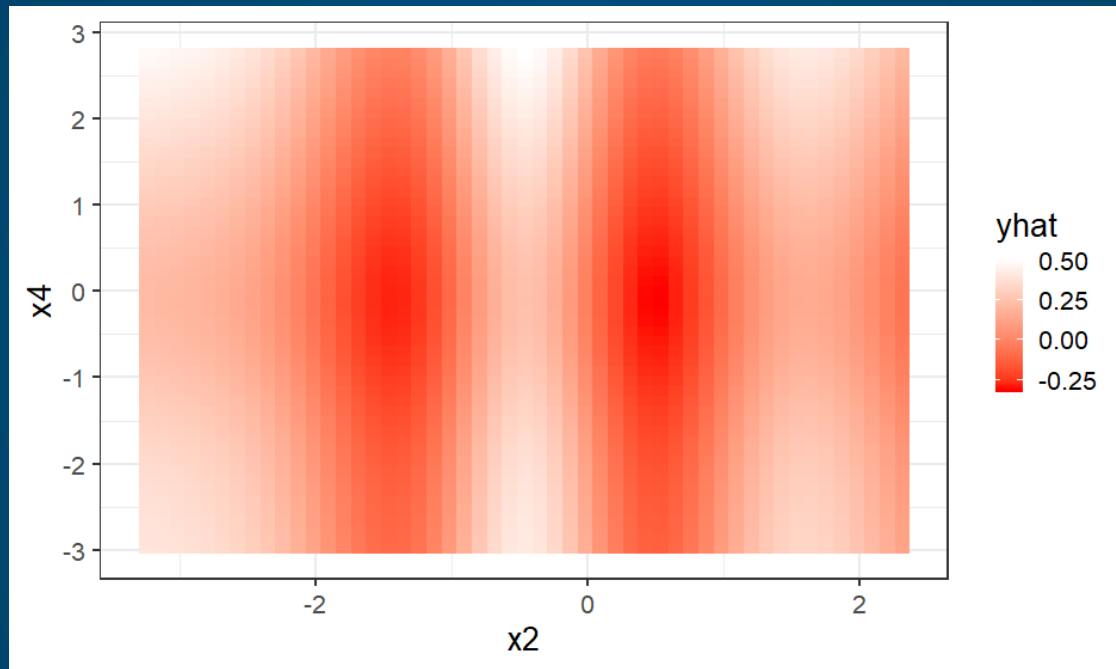
All Variables – Random Forest



Partial Dependence

2-Way

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Locally Interpretable Model-Agnostic Explanations (LIME)

Model Interpretability

LIME

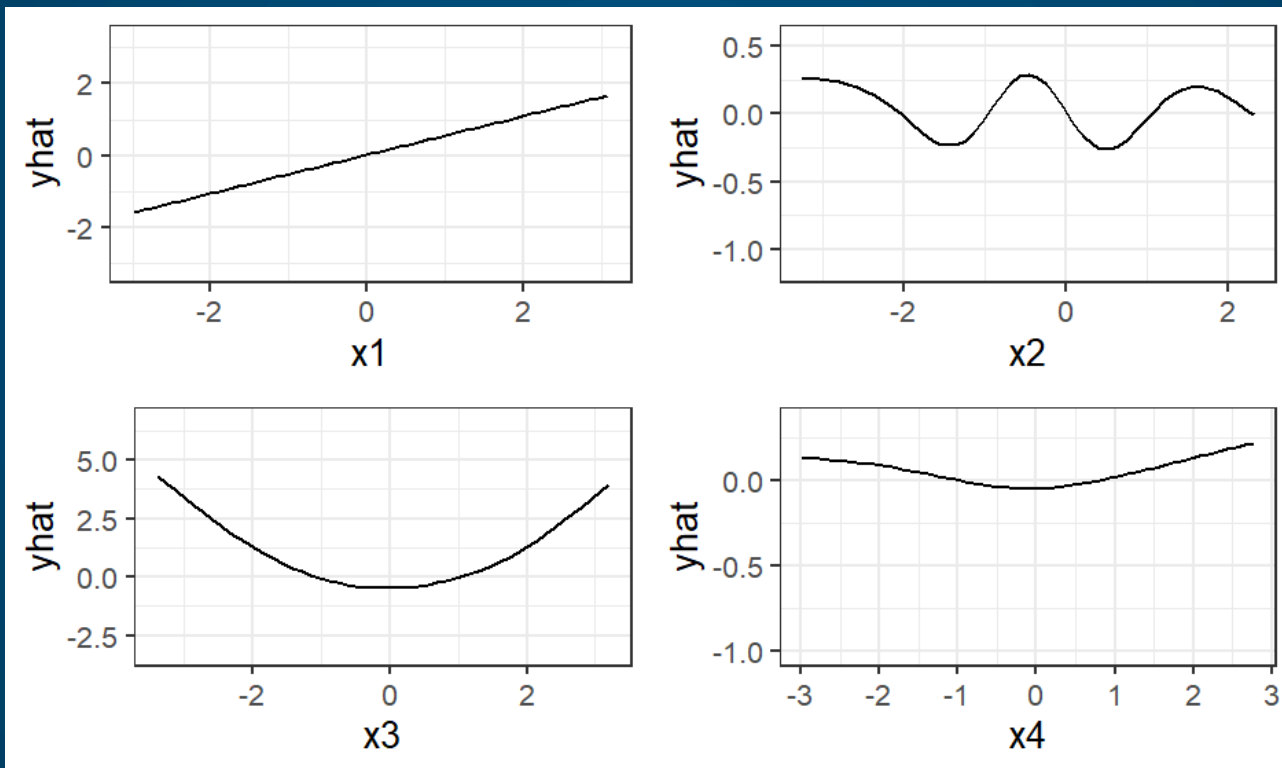
Details

- Evaluates the coefficients of a linear model trained on a model's predictions around an individual observation.
- Steps:
 1. Pick a single observation
 2. Perturb data to generate random observations
 3. Score new observations with model
 4. Weight observations based on their proximity to the observation
 5. Train a linear model on model's predictions
 6. Interpret Model Coefficients

LIME

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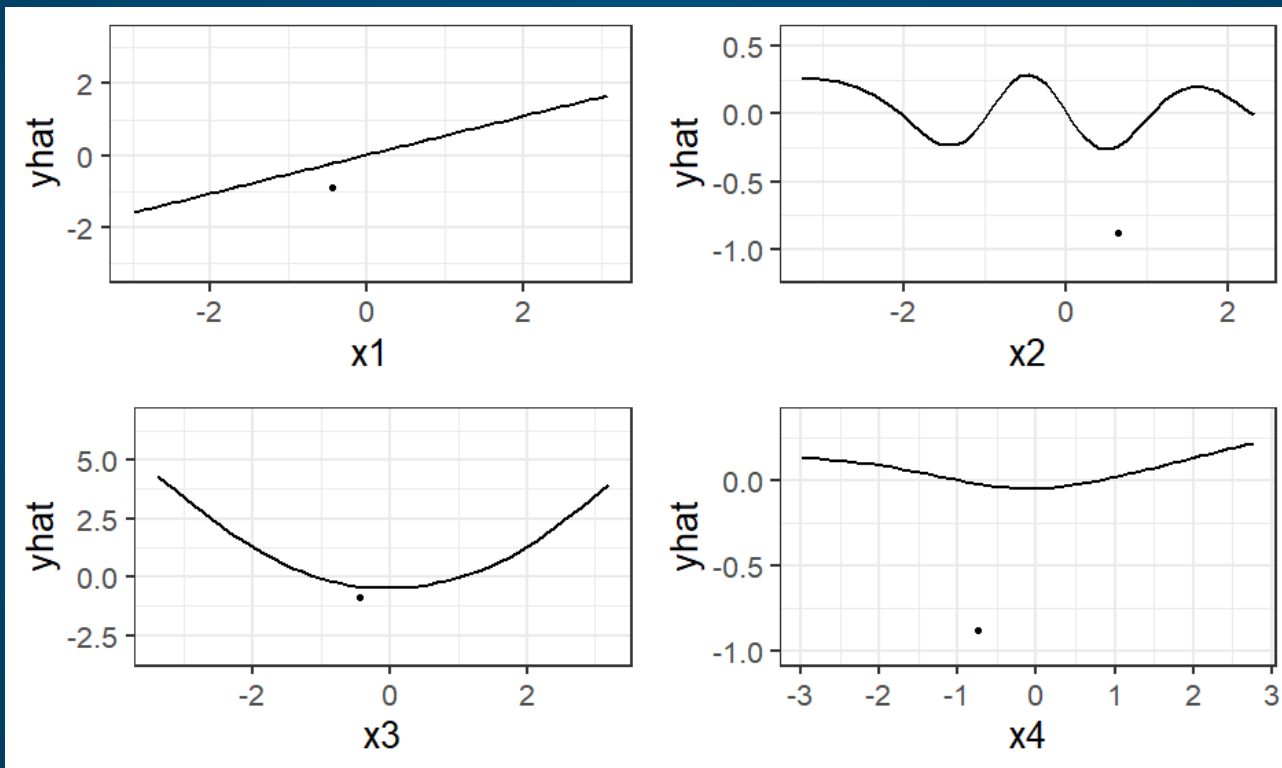
Visualization - Partial Dependence



LIME

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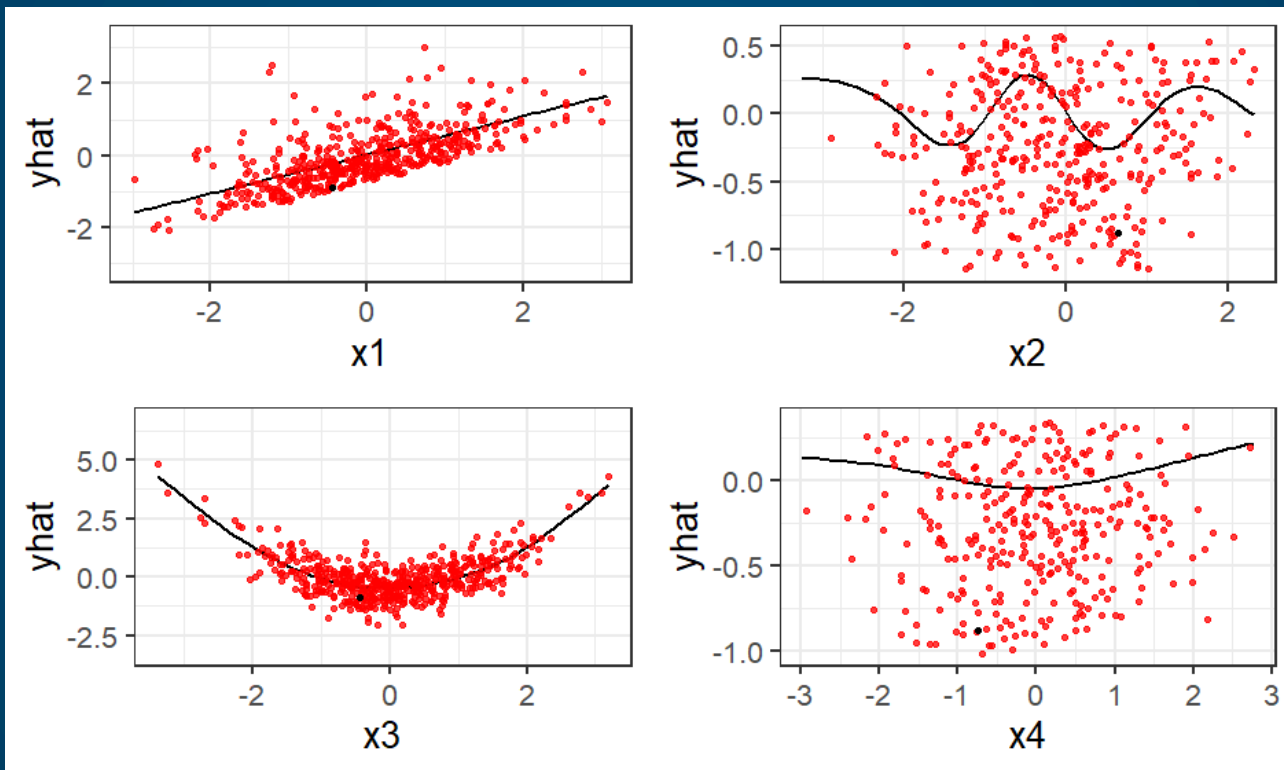
Visualization – Selecting an Observation



LIME

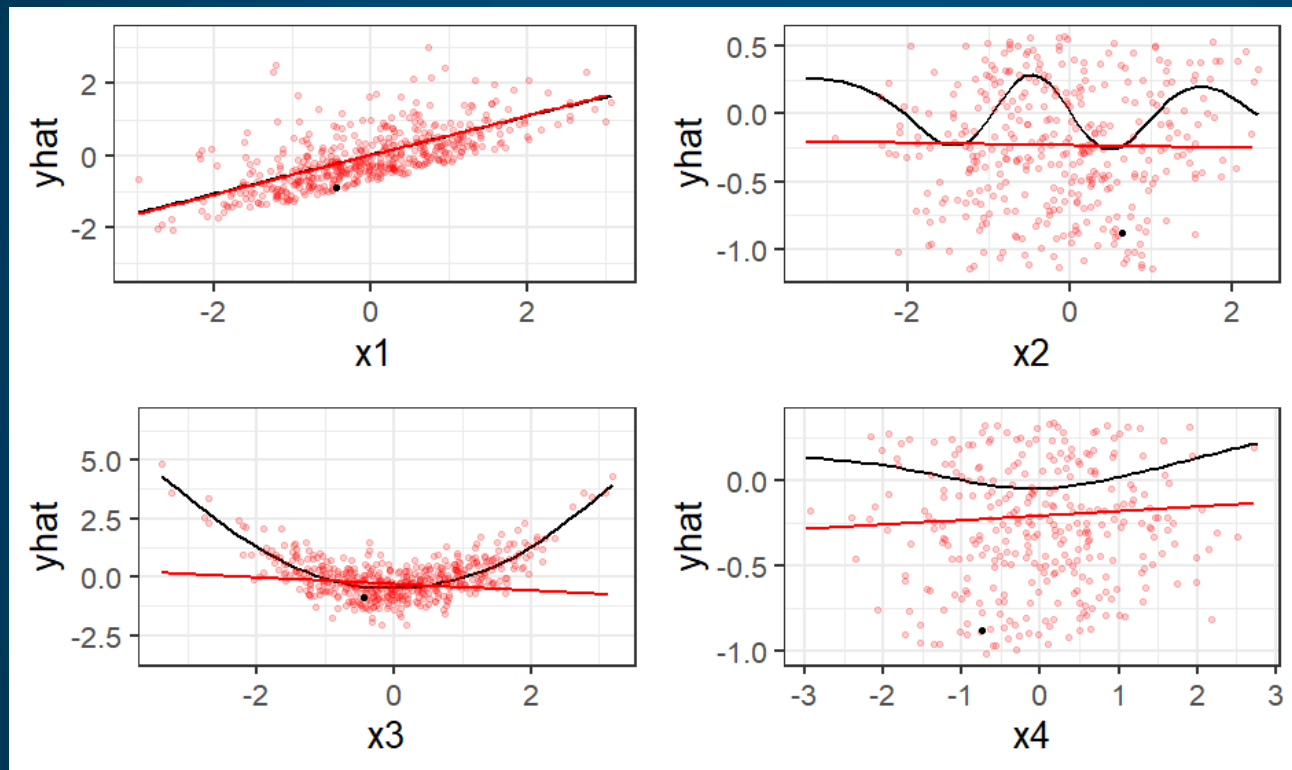
Visualization – Generate Data

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LIME

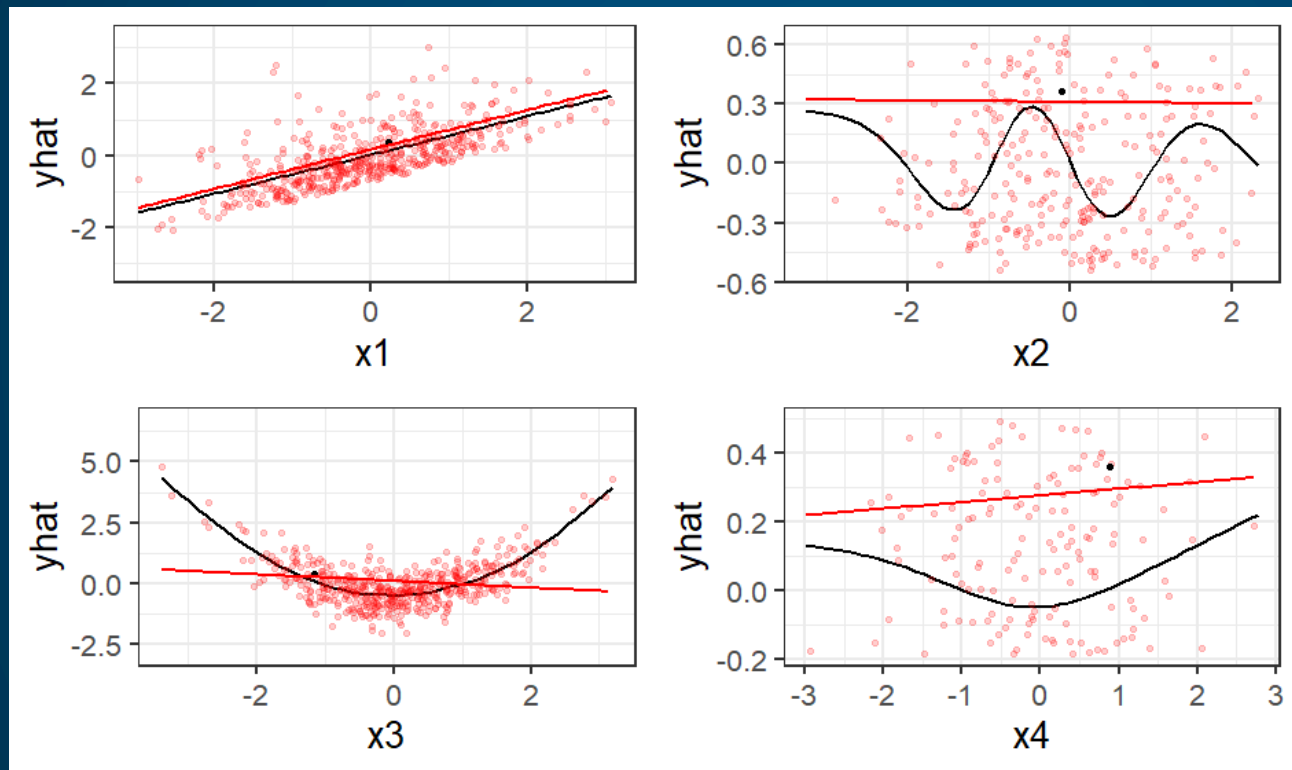
Visualization – Fit Regression



feature	value	weight
x1	-0.444	0.534
x2	0.637	-0.002
x3	-0.440	-0.126
x4	-0.747	0.002

LIME

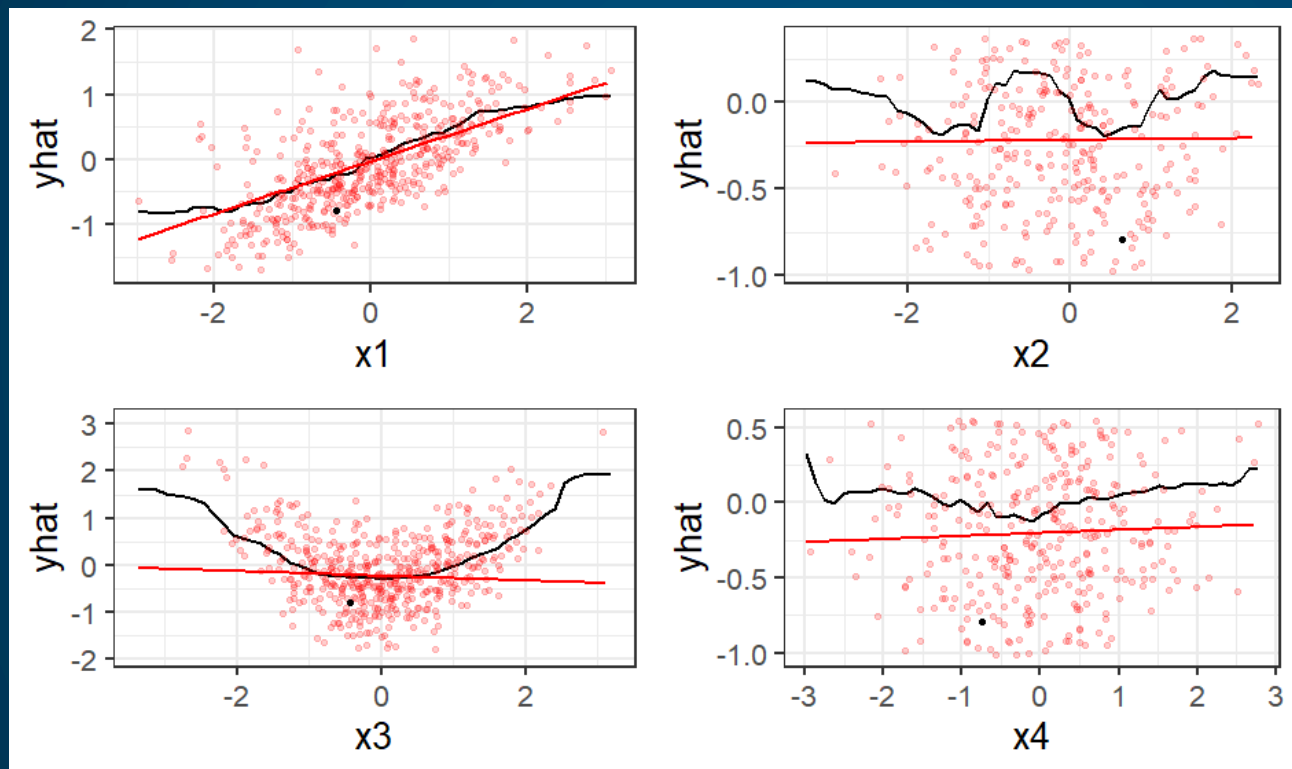
Visualization – Another One



feature	value	weight
x1	0.218	0.526
x2	-0.112	-0.005
x3	-1.163	-0.159
x4	0.879	0.007

LIME

Visualization – Forest Model



feature	value	weight
x1	-0.444	0.408
x2	0.637	0.008
x3	-0.440	-0.058
x4	-0.747	0.018

LIME

List of Warnings

- Automatic Plots return $\text{Beta} * \text{Feature Value}$
- Shapley Values are likely a better fit
- Linear Model Issues
 - Distance metric may need tuning – but how?
 - How many features to select with LASSO?

The Shapley Values

Model Interpretability

Shapley Values

Full Shapley Computation

- Evaluates the effect on a model's prediction of an input variable taking the value it has in the query observation.
- Steps:
 1. Pick a single observation
 2. Retrain the model for every subset of variables
 3. Score each trained model on the observation
 4. Aggregate the scored values to determine the effect of including each variable in the training set, controlling for the other variables.

$O(2^m)$ *training rounds*

Shapley Values

Expected Value Approximation

- Set an entire column in the training set equal to that variable's value in the query, take the average prediction on this perturbed set.
- Reformulate the shapley value calculation to use sets perturbed in this way instead of retraining models.

Shapley Values

Original Dataset and Query

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x1	x2	x3	x4
-0.44363	0.637281	-0.44017	-0.74696
-0.5148	-0.73199	0.969385	1.631754
1.180671	-0.74499	0.509752	1.047877
1.573187	0.264565	0.060567	-1.03539
0.09336	-2.04954	-1.05795	0.634159
0.165143	-0.33239	-0.69795	-2.01421
-0.33853	0.522166	0.364564	-0.63017

Shapley Values

Original Dataset and Query

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x1	x2	x3	x4	yhat
-0.44363	0.637281	-0.44017	-0.74696	-0.88262
-0.5148	-0.73199	0.969385	1.631754	-0.01381
1.180671	-0.74499	0.509752	1.047877	0.50978
1.573187	0.264565	0.060567	-1.03539	0.184054
0.09336	-2.04954	-1.05795	0.634159	0.055351
0.165143	-0.33239	-0.69795	-2.01421	0.177851
-0.33853	0.522166	0.364564	-0.63017	-0.87324

-0.12038

Shapley Values

Effect of Replacing X1

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x1	x2	x3	x4	yhat
-0.44363	0.637281	-0.44017	-0.74696	-0.88262
-0.44363	-0.73199	0.969385	1.631754	0.024448
-0.44363	-0.74499	0.509752	1.047877	-0.36344
-0.44363	0.264565	0.060567	-1.03539	-0.90018
-0.44363	-2.04954	-1.05795	0.634159	-0.23334
-0.44363	-0.33239	-0.69795	-2.01421	-0.14943
-0.44363	0.522166	0.364564	-0.63017	-0.92975

-0.49062

Shapley Values

Effect of Replacing X2

48

x1	x2	x3	x4	yhat
-0.44363	0.637281	-0.44017	-0.74696	-0.88262
-0.5148	0.637281	0.969385	1.631754	-0.444
1.180671	0.637281	0.509752	1.047877	0.088898
1.573187	0.637281	0.060567	-1.03539	0.145174
0.09336	0.637281	-1.05795	0.634159	-0.18586
0.165143	0.637281	-0.69795	-2.01421	-0.32349
-0.33853	0.637281	0.364564	-0.63017	-0.84827

-0.35003

Shapley Values

Effect of Replacing X1 and X2

x1	x2	x3	x4	yhat
-0.44363	0.637281	-0.44017	-0.74696	-0.88262
-0.44363	0.637281	0.969385	1.631754	-0.40574
-0.44363	0.637281	0.509752	1.047877	-0.78432
-0.44363	0.637281	0.060567	-1.03539	-0.93906
-0.44363	0.637281	-1.05795	0.634159	-0.47455
-0.44363	0.637281	-0.69795	-2.01421	-0.65076
-0.44363	0.637281	0.364564	-0.63017	-0.90477

-0.72026

Shapley Values

Interaction of X1 and X2 - GAM

- For the GAM Model, the effects of X1 and X2 are additive!

x1	x2	Δy_{hat}
1	0	-0.3702381
0	1	-0.2296479
1	1	-0.5998859

Shapley Values

Interaction of X1 and X2 – RandomForest

- For the RandomForest Model, the effects of x1 and x2 are not additive!
- The full shapley value computation accounts for this interaction

x1	x2	Δy_{hat}
1	0	-0.3234378
0	1	-0.04472599
1	1	-0.434768

Shapley Values

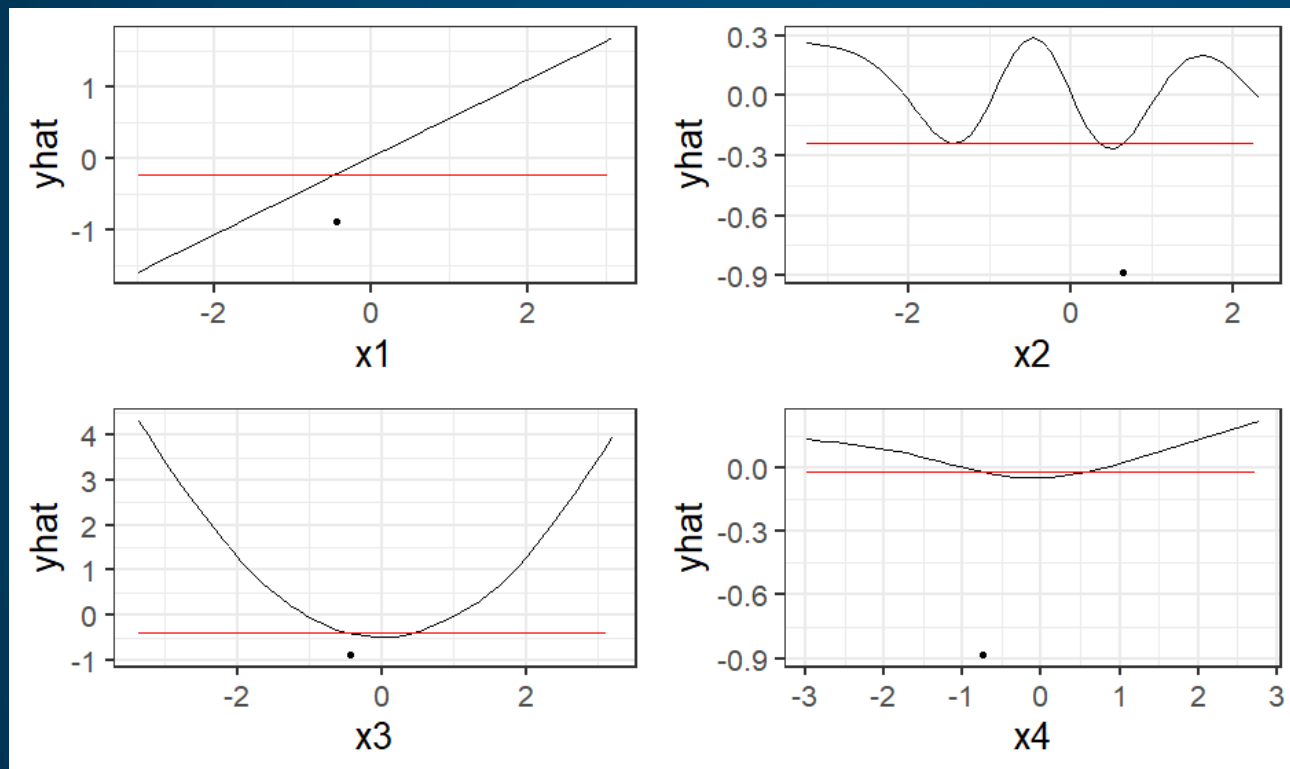
Details

- Evaluates the effect on the model's prediction of varying a variable's value in an entire dataset
- Steps:
 1. Pick a single observation, the “query”
 2. For all subsets of variables, generate a copy of the training set where that set of variables' values are set equal to the values they take in the query
 3. Score new observations with model
 4. Average prediction for each dataset replicate
 5. Aggregate
 6. Plot

Shapley Values

Plots - GAM

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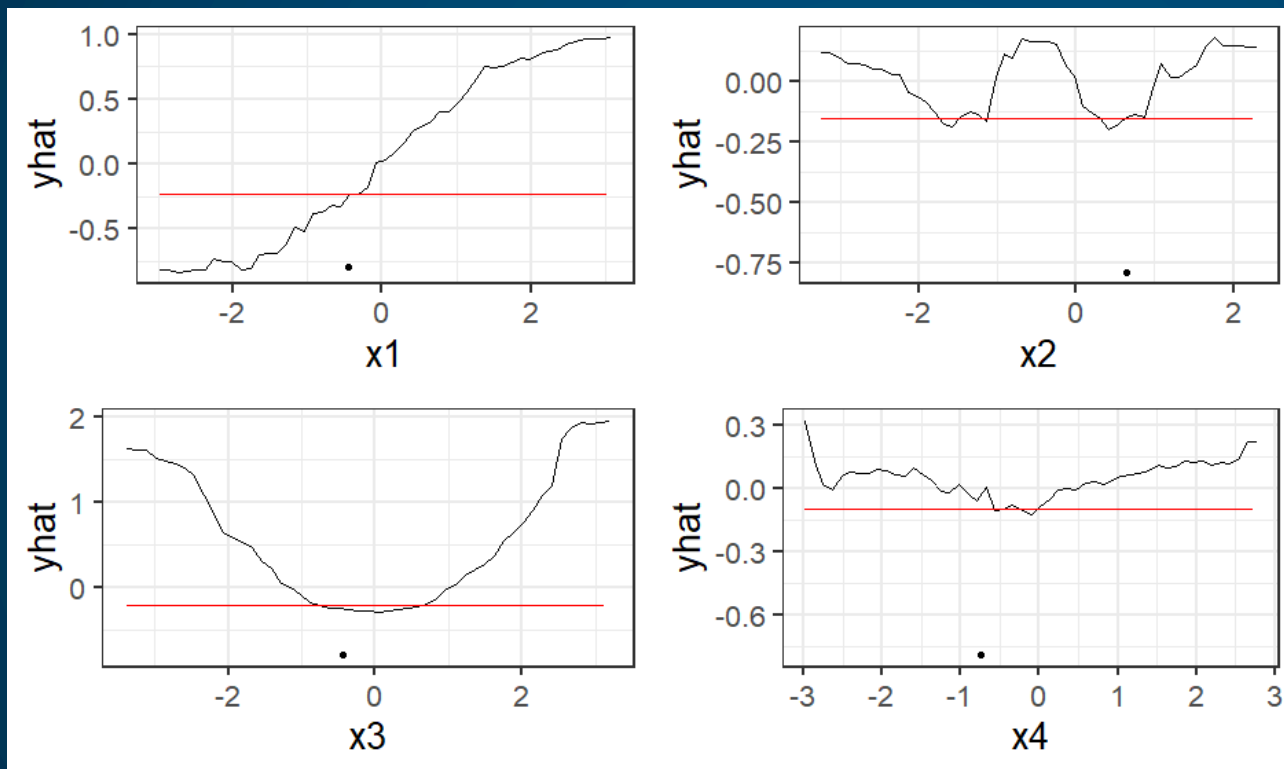


feature	phi
x1	-0.18923
x2	-0.24207
x3	-0.45175
x4	-0.02025

Shapley Values

Plots – Random Forest

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feature	phi
x1	-0.2622
x2	-0.17979
x3	-0.26623
x4	-0.0918

Shapley Values

Approximation Methods

- KernelSHAP (or just SHAP) – approximates Shapley Values w/ Linear Model
- Shapley Sampling Values – estimates Shapley Values by aggregating perturbed samples
- HyperSHAP (SAS Method) – approximates Shapley Values by reducing the number of variable sets used